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Ana-Paula Correia, Sean Hickey & Fan Xu

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Realizing the possibilities of the large language models: Strategies for prompt engineering in educational inquiries

Ana-Paula Correia ^a, Sean Hickey^b, and Fan Xu^{b*}

^aCenter on Education and Training for Employment, Educational Studies Department, College of Education and Human Ecology, The Ohio State University, Columbus, Ohio, USA; ^bCenter on Education and Training for Employment, College of Education and Human Ecology, The Ohio State University, Columbus, Ohio, USA

ABSTRACT

This article examines the potential of Large Language Models (LLMs) to transform educational inquiries, emphasizing effective prompt engineering strategies. It begins by introducing LLMs and applications of generative AI tools in education. The article explores the fundamental principles of prompt engineering and LLMs, outlining strategies for educators to interact effectively with these models. It emphasizes the vital role of teacher-LLM interactions, highlighting the need for carefully crafted prompts to elicit high quality and pedagogically valuable responses. We discuss chain-of-thought prompting as a method for promoting deeper reasoning in LLM outputs. We offer other practical strategies for effective prompt engineering, providing actionable insights for teachers to enhance their professional practices. We also address ethical and practical considerations, including bias in AI responses and the importance of teacher autonomy and professional judgment.

Introduction

For over a decade, predictive artificial intelligence systems have suggested words and phrases on mobile phone messaging applications, aided with grammar and writing style within office applications, and acted as digital assistants in smart speakers and mobile devices (Kutela et al., 2023). These tools use advanced statistical models to interpret input and suggest output based on large libraries of example data. They are powered by a predictive algorithm that examines input and is backed by enormous, curated databases of possible responses to provide the desired output. These predictive models, however, are limited by the training data they have received; they cannot learn on their own. In other words, “if a user asks the virtual assistant to do something that is not in its code, the bot simply says it can’t help” (Chen et al., 2023, para. 12).

Generative artificial intelligence (AI) systems, on the other hand, follow a different approach. Unlike their predecessors, generative systems are not limited to categorizing input and using predictive statistics to match that input to options found in a library of

CONTACT Ana-Paula Correia  correia.12@osu.edu  Center on Education and Training for Employment, Educational Studies Department, The Ohio State University, 1900 Kenny Road, Columbus, OH 43210, USA

*Fan Xu’s current affiliation: Department of Teacher Education, Educational Leadership and Policy (TEELP), College of Education and Human Sciences, University of New Mexico (UNM)

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preexisting output. Instead, they can generate output that, while often similar to, using components of, or mimicking the examples on which it has been trained, is entirely novel, unique to the human input received (Lund & Wang, 2023). These systems are named large language models (LLMs) because they are trained on vast amounts of textual data and have many parameters that enable them to understand and generate human-like language.

An example is the generative pretrained transformer (GPT), of particular interest since the release of OpenAI's ChatGPT systems at the end of 2022, one such model in which input is processed by an algorithm that has been trained with large amounts of data to generate a desired response (Lund & Wang, 2023; Ortiz, 2023). ChatGPT relies on a generative AI model to perform specialized language processing tasks, including carrying out conversations and generating text-based content. The “chat” at the beginning of “ChatGPT” refers to the system's ability and its chatbot-type interface, which can interpret conversational text, analyze it, and provide a human-like response in return.

Educators need to develop strategies for creating effective written instructions or prompts when interacting with GPT-based applications, a process known as prompt engineering. In the context of interacting with LLMs, a *prompt* is a user-provided text input that initiates the model's generation process and defines its output. It acts as a directive or question that guides the model in producing a relevant and contextual response.

Prompt engineering in educational settings

Crafting prompts poses a challenge for teachers and educators due to the inherently intricate nature of educational inquiries and the AI systems' inaccuracies, also known as “hallucinations” (Salvagno et al., 2023). By utilizing proven strategies for prompt engineering, teachers can interact with these systems efficiently to maximize the potential of LLMs to create personalized learning experiences to meet students' individual needs. Empirical research (e.g., X. Wang et al., 2022) has demonstrated that utilizing Chain-of-Thought prompts enhances the performance of LLMs across a wide range of intricate reasoning tasks, as well as other strategies discussed in existing research.

Understanding prompt engineering and LLMs

Generative AI tools powered by LLMs are increasingly integrated into educational spaces. Instructors need to not only familiarize themselves with these tools but also develop best practices for their use and inclusion in lesson planning, assignments, and classroom activities. They are of particular interest in education because of their ability to understand input and generate output using natural human language rather than requiring structured queries or computer programming knowledge (Jiang et al., 2020). Having been trained on millions of written-word samples, these tools can generate sophisticated novel content for a variety of purposes. For example, LLMs can provide student feedback or automate grading by analyzing student work and identifying errors, strengths, or weaknesses (Kasneci et al., 2023). LLMs have the potential for use in a wide array of instructional roles in the classroom, including content provider, evaluator, or teaching assistant (Jeon & Lee, 2023).

That is not to suggest that LLM use in the classroom is without drawbacks. Kasneci et al. (2023) note that LLMs have the potential to perpetuate stereotypes, preconceptions, and biases found in the training data. Further, LLMs may fail to fully adapt to the diverse needs of students,

or they may require additional training to be capable of providing an adequate level of personalization to meet each student's unique needs, idiosyncratic circumstances, and learning contexts. Jeon and Lee (2023) note that the effectiveness of LLMs in educational contexts is highly dependent on the teacher's ability to formulate high quality prompts, meaning educators need to develop specific AI-related competencies to maximize the potential of LLMs for instruction.

Cain (2024) suggests that skill in prompt engineering is essential for educators who hope to maximize the potential of LLMs to personalize students' learning experiences, describing prompt engineering as an iterative process of "shaping the AI's responses" (p. 49) by first thoughtfully designing the initial prompt and then by carefully refining it until the desired output is produced. These 2 phases — creation and revision — are reminiscent of the differentiation by Google Cloud (2024) between prompt *design* and prompt *engineering*, the former being the creative work of initially developing the prompt and the latter being an iterative, more technical honing process. This is echoed by Correia and Hickey (2024), who emphasize the inherent creativity embedded in designing an effective prompt. Beyond creativity, Cain (2024) highlights deep content knowledge and critical thinking skills as two additional components of effective, prompt writing, suggesting that educators need to provide the necessary content knowledge to successfully guide the LLM and effectively evaluate the response to detect inaccuracies and judge its suitability for instruction.

Teacher-LLM interactions

Teachers skilled in prompt engineering can extract more nuanced and specific feedback from generative AI tools to support student learning. Educators also need to model and teach ethical considerations when using AI, including understanding the biases inherent in AI responses based on the prompts given, as reported below.

Educators as "prompt engineers"

Teachers are embracing a new role as prompt engineers in the AI-evolving landscape. This changing role aligns with the 2024 UNESCO "AI Competency Framework for Teachers" (Miao & Cukurova, 2024), which emphasizes that teachers must develop competencies not only in the technical application of AI tools but also in ethical decision-making, pedagogical integration, and ongoing professional learning. The framework underscores a human-centered mind-set, promoting teacher agency and accountability as essential in the age of AI.

Along these lines, Isemonger (2023) draws a parallel between effective communication with LLMs and the specialized information gathering skills of professionals such as lawyers, police investigators, and talk-show hosts. This analogy highlights that skillful interaction with LLMs is a critical skill comparable to effective human communication. For teachers, mastering the principles of prompt engineering enhances their understanding of these models' capabilities and limitations. This knowledge permits them to extract nuanced, targeted outputs from generative AI tools, enriching both the teaching and the learning experiences.

According to a recent Pew Research Center survey (Lin et al., 2024), many public K-12 teachers report insufficient time during work hours to complete their responsibilities. Generative AI tools can help streamline administrative tasks and automate routine processes, improving productivity and allowing teachers to dedicate more energy to higher-priority goals, such as meaningful interactions with students and families. Simple tasks can often be achieved with straightforward prompts. For example, a teacher can provide key

information, such as the date, time, location, and purpose of a parent-teacher conference, and an LLM like ChatGPT can generate a well-structured e-mail invitation.

However, more cognitively demanding teaching tasks require well-engineered prompts. Teachers can design AI-integrated curricula in curriculum development by leveraging LLMs to search for relevant resources, generate engaging learning activities, and draft syllabi. AI tools can also assist in assessing student work and providing targeted feedback. While ChatGPT and similar tools are valuable for such tasks, the teacher's input remains central to creating meaningful and effective educational experiences. The thought process behind a teacher's curriculum design reflects their expertise, pedagogical philosophy, and deep understanding of their student's needs and learning contexts, making their work uniquely impactful. Simple prompts risk stripping away teachers' intentions and creativity, resulting in over-generic or impersonal educational solutions.

Chain-of-thought prompting

A common concern regarding using LLMs in education is the significant inaccuracy and uncertainty of their responses. This phenomenon, known as AI "hallucinations," refers to the generation of incorrect factual content and the fabrication of seemingly valid but non-existent bibliographic references (Salvagno et al., 2023). While it is widely acknowledged that the capabilities of LLMs are far from perfect, inaccuracies in education and other high-stakes domains, such as medicine, can lead to more severe consequences compared to other fields.

Direct prompts often result in incomplete or incorrect responses, especially for tasks requiring multi-step reasoning, such as solving mathematical problems or generating code (C. Wang et al., 2024). Yet, research suggests that crafting sophisticated prompts is an effective way to tap into the latent potential of LLMs without modifying their underlying parameters (C. Wang et al., 2024; Zhou et al., 2022). Advanced prompt engineering techniques, such as in-context learning (Brown et al., 2020), chain-of-thought prompting (X. Wang et al., 2022), and self-consistency (X. Wang et al., 2022), have been proposed to enhance the reliability and accuracy of these models when handling complex tasks.

Additional concerns when using LLMs is their tendency to skip critical reasoning steps, providing students with direct answers that lack transparency. However, it is the thinking process behind the answer that students need to develop when engaging with AI-powered tutoring. By employing techniques such as chain-of-thought prompting, educators can guide LLMs to outline their reasoning step-by-step, making the process of how the models arrive at an answer more transparent and accessible to students (X. Wang et al., 2022; Wei et al., 2022; Zhang et al., 2022).

Chain-of-thought prompting is a widely used advanced prompting method that guides LLMs to produce detailed, step-by-step reasoning in their responses to a task (X. Wang et al., 2022; Wei et al., 2022; Zhang et al., 2022). By explicitly mimicking human "think-aloud" processes, this technique guides the model to break down intricate problems into smaller, manageable steps, leading to more accurate and coherent outputs. This approach not only improves the quality of the responses but also provides students with a visualized reasoning process that helps them understand correct problem-solving methods, identify errors in their previous approaches, and refine their critical thinking skills.

By integrating chain-of-thought prompting into educational practices, educators can transform generative AI tools into powerful resources for scaffolding learning. When guided with well-structured prompts, these tools not only assist in solving problems but

also foster a deeper understanding of reasoning processes, equipping students with critical thinking skills essential for lifelong learning.

The advantages of chain-of-thought prompting for educators

Wei et al. (2022) give an example that shows rather than simply asking an LLM to solve a math problem, educators can prompt the model to show its reasoning process step-by-step. This approach applies beyond math to tasks such as analyzing complex symbolic logic or interpreting educational data. Research by McClure et al. (2024) highlights the effectiveness of chain-of-thought prompting in enhancing LLMs' ability to assess cognitive engagement levels in student responses. Their study demonstrated improved comprehension and analytical capabilities in LLMs by employing a structured 7-step Chain-of-Thought sequence, leading to more accurate and insightful assessments (McClure et al., 2024).

Building on this, chain-of-thought prompting enables educators to guide LLMs through intricate reasoning tasks by chunking them into intermediate steps. Figure 1 shows an example of its potential application in curriculum design. Chain-of-thought prompting enables LLMs to design effective curriculum activities by incorporating student features, knowledge content and objectives, and structured reasoning steps.

Well-crafted prompts not only guide the LLM in performing tasks but also capture the teacher's reasoning behind specific learning activities or assessment strategies. This process permits teachers to document their thought processes, critically evaluate AI outputs, identify areas for improvement, and make necessary adjustments aligned with their educational goals and values. By combining their own insights with the capabilities of LLMs, the teachers can ensure that LLMs complement, rather than replace, their professional judgment. This also allows teachers to maintain control over their classrooms' pedagogical direction, preventing over-reliance on AI-generated outputs. Moreover, teachers can use advanced prompts to tailor AI-generated content to address diverse student needs, ensuring inclusivity and accessibility. Additionally, they can refine AI-generated assessments to ensure the feedback is accurate, relevant, and constructive, with broader implications for educational equity and fairness.

By creating detailed prompts for LLMs to generate high-quality responses, educators provide students with structured guidance that enhances their understanding of complex concepts (Heston & Khun, 2023). Advanced prompts help students navigate learning tasks with greater clarity and encourage them to explore topics from diverse perspectives, whether from humans or machines. Teachers who are adept in prompt engineering can guide students in formulating better questions for AI and critically evaluating its answers. Mastering prompt engineering skills enables educators to model and teach ethical AI use, including the ability to recognize and address biases in AI responses that may result from poorly constructed prompts. Ultimately, educators with prompt engineering skills not only make effective use of AI tools but also prepare students to navigate the AI-driven future safely and responsibly.

Strategies for effective prompt engineering in education

The capabilities of LLM models can be attributed in part to an attention mechanism that allows them to focus on specific components of the training data and establish connections across extended text segments (Elkins & Chun, 2020; Lund & Wang, 2023). For educators to take

Standard Prompting

Model Input

Q: Create a 10-minute class activity to teach high school students about mountain fire.

A: Collaborate effectively in teams to propose solutions for mitigating fire risks and managing fire emergencies.

Q: Create a 10-minute class activity to teach middle school students about gravity.

Model Output

A: Collaborate in small groups to design a simple experiment that demonstrates gravity in action.

Chain-of-Thought Prompting

Model Input

Q: Create a 10-minute class activity to teach high school students about mountain fires.

A: Step 1 – Understanding my students: High school students possess stronger communication and critical thinking skills, making them capable of engaging in discussions and problem-solving.

Step 2 – Understanding my learning content: Mountain fire education includes societal issues, environmental responsibility, and the importance of prevention and emergency response.

Step 3 – Decide my learning activity: Based on the students' capabilities and the nature of the topic, an effective activity is: Collaborate in teams to propose solutions for mitigating fire risks and managing fire emergencies.

Q: Create a 10-minute class activity to teach middle school students about gravity.

Model Output

A: Step 1 – Understanding my students: Middle schoolers are curious and enjoy hands-on activities. They benefit from tangible demonstrations and straightforward explanations of abstract concepts.

Step 2 – Understanding my learning content: Gravity is a fundamental force that pulls objects toward the center of the Earth. Teaching gravity involves demonstrating how it works in real life and why it's essential for motion, balance, and planetary systems.

Step 3 – Decide my learning activity: Based on the characteristics of my students and the knowledge point, a good activity is: Experimenting with gravity by comparing the motion of falling objects.

Figure 1. Chain-of-thought prompting guides curriculum design through reasoning steps. *Note.* Chain-of-thought reasoning processes are organized step-by-step. Q = Question; A = Answer.

advantage of these connections and make the most of an LLM's capabilities, however, they need to consider the wording of their prompts and select the right words and phrases to achieve the desired output. Veletsianos et al. (2023) liken LLM prompts to "magic spells," suggesting that precision is needed for the AI tool to respond correctly. According to Ortiz (2023), LLM users find this precision by rephrasing their requests multiple times to get the output they want. Thunström and Steingrímsson (Transformer et al., 2022) similarly argue that while straightforward and general prompts have the potential to yield high-quality results, trial and error is more often needed to elicit the ideal response from an LLM-based AI tool.

It stands to reason, then, that the educators who successfully use LLMs and receive the desired response are those who have developed, practiced, and refined their skills related to creating AI prompts. By utilizing proven strategies for prompt engineering, teachers can interact with generative AI tools efficiently to maximize the potential of LLMs to create personalized learning experiences. Effective prompt engineering allows educators to generate desired information or resources from AI more quickly, saving time that can be spent on other, more impactful educational tasks. The content of the prompt can range from a simple command or query to a more complex set of instructions or a scenario description. The model's response is contingent on the prompt's wording, and hence, the quality and specificity of the prompt can significantly influence the accuracy, relevance, and usefulness of the model's output. Park and Choo (2024) enumerate the essential components of effective prompt engineering using the acronym PARTS: Persona, Aim, Recipients, Theme, and Structure. By providing a persona, educators can identify to the LLM the "role" the AI tool needs to take on, defining critical context for the model. The *persona* can be an instructional or educational role, such as a special education teacher, or it can embed the LLM in the instructional content, such as the instruction line, "Act as if you are Martin Luther King, Jr" (Park & Choo, 2024, p. 3). The *aim*, *recipients*, and *structure* describe the essential work to be done by the LLM by defining the task (e.g., crafting an e-mail or writing a story), the audience (e.g., fourth-graders or first-year college students), and the format of the output (e.g., length, language, or reading level).

While PARTS covers the initial design of the prompt, Park and Choo (2024) tackle the iterative engineering process with 2 additional acronyms: CLEAR and REFINE. The former provides a framework for ensuring the LLM will understand the request by encouraging educators to be Concise in their instructions, Logical in their phrasing, Explicit with what they are requesting, Adaptive to the needs of their student audience, and Restrictive, framing their request within a specific content domain. Following this process of revising and honing AI prompts to ensure the best possible results, the first 2 steps of Park and Choo's REFINE acronym provide suggestions on how the initial prompt might be improved in subsequent iterative steps, such as by Rephrasing terms or Experimenting with examples and syntax. After these first 2 steps, it is essential for users to remember that LLM use often involves a back-and-forth, dialogue-style interaction. Unlike a simple Google search, which begins with search terms and ends with a page of results, LLMs utilize a conversational format to allow for more of a discussion for refining results. To make this possible and further reduce the need for a high degree of precision in the initial prompt, Reinforcement Learning from Human Feedback was used in the development of LLMs, improving their conversational interactions by training the model with examples of both sides of a conversation (Ortiz, 2023). The remaining 4 steps of the REFINE acronym encourage educators to take advantage of this conversational nature of generative AI tools. Park and

Choo (2024) suggest using the conversation with the LLM as a Feedback loop, Inquiring of the LLM what information it might need to produce the desired results, Navigating the chatbot interface to compare multiple outputs or responses, and Evaluating each output for both accuracy and its applicability for classroom discourse.

The literature provides additional effective prompting strategies that can aid teachers in leveraging generative AI tools. For example, White et al. (2023) present a comprehensive prompt catalog designed to enhance prompt engineering with ChatGPT. Similarly, Correia and Hickey (2024) provide 5 best practices for prompt design, offering valuable guidance for educators seeking to optimize their use of AI tools in the classroom.

Crosthwaite et al. (2024) found that teachers who were able to craft effective prompts experienced positive educational outcomes among their students, suggesting that the most effective prompts included clear instructions and contained both the context in which the output will be used and the definition of the pedagogical goals of the LLM output. For example, by providing the LLM with a content outline and key learning objectives, teachers were successful at having generative AI tools to write questions and multiple-choice test items that could be used for both formative and summative assessments in the classroom. Similarly, by clearly defining the unique needs of individual students or groups of learners, teachers were able to differentiate existing instructional materials, adjusting language complexity based on the various levels of proficiency among the students or incorporating culturally relevant examples into the content. Molina et al. (2024) similarly found that selecting the specific details to include within the prompt was an essential skill for instructors in getting the desired output from generative AI tools. For example, by providing specific information about students' learning levels, preferences, and reading abilities, educators could generate activities appropriate for different skill levels. By giving existing lesson plans and indicating pedagogical strategies in the prompt, Molina et al. (2024) found that LLMs could redesign lessons to incorporate active learning techniques or specific instructional strategies (e.g., think-pair-share, real-world case studies) to encourage student engagement.

Teachers are not the only ones in the classroom that can benefit from developing effective prompt engineering skills. Engaging with generative AI as part of classroom activities can provide significant educational benefits for students, including improved problem-solving skills, better classroom engagement, and a greater technical understanding of the capabilities and limitations of LLMs and generative AI tools (Theophilou et al., 2023). Through trial and error, students interacting with LLMs have been shown to discover the need for clear and concise prompts quickly. In studying high school students learning to use AI in the classroom, Theophilou et al. (2023) noted student frustration related to the need to clarify and refine their prompts, highlighting a need for prompt-engineering education and a potential limitation for using generative AI tools with younger students. The researchers also noted that some students exhibited some level of apprehension about using AI but that these fears generally diminished over time with more exposure to the tools and more experience creating prompts, highlighting another potential benefit for both teachers and students in learning how to design effective AI prompts.

There is no "right answer" when it comes to creating effective prompts for use with LLMs and generative artificial intelligence. The teachers examined by Crosthwaite et al. (2024), for example, used a variety of different frameworks and formats for crafting their prompts. Although these frameworks varied, none were shown to be universally superior. Instead,

their utility depended on how well they aligned with the instructional intentions of the educator. Researchers emphasized the importance of clarity, context, and iterative refinement in designing and engineering prompts to yield desired results from LLMs. Thus, prompt frameworks should be viewed not as rigid formulas but as scaffolds to help educators articulate their educational goals and tailor prompts accordingly. Educators should perhaps focus less on a specific framework or model when creating prompts and more on the thought processes behind the prompt, being sure to clearly and concisely indicate the overall pedagogical or instructional goal, the format of the desired output, and the intended audience. This notion aligns with the GPEI methodology proposed by Velásquez-Henao et al. (2023), which encourages LLM users first to consider their overall Goals for using generative AI before writing an initial Prompt, followed by Evaluation of the prompt's efficacy and, finally, Iteration on the prompt to refine the results. This broad framework for LLM use treats prompt design less like a formula with variables to be filled in and more as a problem solving strategy in which much of the initial planning work for creating the prompt exists outside of the AI chatbot interface as instructors first consider their instructional and pedagogical goals. Such flexibility emphasizes that it is not the framework itself that ensures success but how effectively the framework serves the instructional purpose dictated by the educator.

For more complex uses of generative AI in the classroom, educators may need to create more detailed and complicated prompts, further highlighting the need for advanced planning and consideration of pedagogical or instructional goals *prior* to engaging with AI tools. For example, Wan and Chen (2024) describe a process for training an LLM to provide written feedback for student writing samples. The process starts by gathering data related to student writing, including a wide variety of writing samples that will eventually be given to the LLM for analysis. Then, they create an initial prompt, preparing the generative AI tool to give student feedback by providing it with the writing assignment or instructions, a rubric for grading the assignment, and other student-facing guidelines related to the assignment. They then pair this prompt with a small number of example essays, plus high quality instructor feedback for each of the essays, giving the LLM examples of the format and style of the feedback they expect it to generate. Finally, to refine LLM's responses, they use an iterative process in which instructors identify strengths and weaknesses in the AI output/answer, provide additional examples of student work and high quality, human generated feedback, and refine the initial prompt as needed to get the best output. Wan and Chen (2024) found that students found the AI-generated writing feedback to be useful and comparable to the feedback they had previously received from human instructors. In some cases, students noted that the feedback received from the AI tools was more detailed than what they typically received from human instructors, noting that the AI was effective at identifying passages where students had reached a valid conclusion but failed to provide sufficient reasoning. While this use case involves a lengthy training process, the potential for both time savings in future grading efforts and providing personalized feedback for each student highlights the value of LLM use in the classroom.

The most effective strategies for writing prompts for classroom use require planning on the part of the teacher prior to engagement with the AI tools. Educators need to identify the underlying goals for using LLMs, considering both the value to the instructor (e.g., time savings, creation of instructional materials) and educational value for the student (e.g., personalized learning materials or feedback). While prompt engineering frameworks and

models propose a variety of steps to take and information to include, their shared elements — such as specifying the prompt/request concisely, providing relevant context, offering examples, and iterating based on responses — are not rigid mandates but adaptable strategies (Kulkarni & Tupsakhare, 2024). With advance planning and by implementing these basic strategies, educators can optimize the quality of AI-generated responses and achieve results that will positively impact instruction.

To facilitate strategic engagement with AI tools, Miao and Cukurova (2024) framework recommends that nations and educational institutions create localized AI competency frameworks tailored to specific contexts. These frameworks should be aligned with existing professional standards and provide structured development across three levels: Acquire, Deepen, and Create. Further, teacher education programs should incorporate formal training modules focused on AI prompt engineering, AI ethics, and the pedagogical use of generative tools, spanning pre-service, in-service, and lifelong learning pathways.

Ethical and practical considerations in prompt engineering

As educators increasingly incorporate LLMs like ChatGPT into their teaching practices, it is essential to consider both the ethical and practical implications of prompt engineering. While these AI tools offer innovative ways to enhance learning, they also present challenges related to bias in AI responses, teacher autonomy, and the accuracy of generated content. To navigate these challenges effectively, educators can adopt specific strategies for ethical and practical prompt engineering. These strategies aim to reduce bias, preserve teacher autonomy, and ensure content accuracy. Before outlining these approaches, it is crucial to understand the rationale behind their necessity.

One significant concern is the potential for biases in LLM-generated responses. These models are trained on vast datasets that may contain inherent biases, leading to outputs/answers that reflect societal prejudices. This raises issues regarding the neutrality and inclusiveness of AI-generated educational materials. The phrasing of prompts can significantly impact the AI's output, either mitigating or exacerbating these biases. For example, prompts that explicitly request balanced viewpoints or that ask the model to consider diverse sources can help minimize biased outputs. Conversely, poorly designed prompts may unintentionally reinforce stereotypes or discriminatory views.

Moreover, the spread of misinformation and misrepresentation of research data are notable limitations when interacting with generative AI tools. Sallam (2023) highlights fears surrounding these issues, emphasizing the need for educators to be vigilant. The convincingly worded but potentially inaccurate responses from ChatGPT can mislead both teachers and students. As Ortiz (2023) points out, “critics argue that these tools are just very good at putting words into an order that makes sense from a statistical point of view, but they cannot understand the meaning or know whether the statements it makes are correct” (para. 32). This underscores the importance of critically assessing AI-generated content for accuracy and bias. In line with Miao and Cukurova (2024) recommendations, equipping educators with AI-related competencies also supports long-term professional growth. These competencies include the ability to use AI tools effectively in pedagogy, apply ethical principles, and maintain a critical stance when evaluating AI outputs. Teacher training programs should embed these dimensions,

ensuring progression across the stages of acquiring, deepening, and creating AI-integrated teaching and learning strategies.

Educators can employ several practical strategies to promote responsible use of LLMs in the classroom, as follows:

- Craft prompts that encourage multiple perspectives, such as asking the AI to compare contrasting viewpoints or to utilize diverse sources.
- Use counter-biasing language, such as explicitly instructing the AI to “avoid cultural or gender bias” or to “provide inclusive examples.”
- Critically evaluate AI outputs, cross-referencing them with reliable sources to ensure accuracy before using them in instruction or assigning them to students.
- Encourage students to verify AI-generated information and reflect on its credibility, which fosters digital literacy and critical thinking skills.

Effective prompt engineering supports teacher autonomy and professional judgment by enabling educators to generate tailored resources and solutions that align with their pedagogical goals. By mastering the art of crafting prompts, teachers can guide LLMs to produce content that meets the specific needs of their students. This empowerment enhances teacher agency, allowing for more personalized and effective teaching strategies. However, Cotton et al. (2023) echo concerns about academic dishonesty, noting that “it can be difficult to distinguish between a student’s own writing and the responses generated by a chatbot application,” making “it difficult to adequately assess the student’s understanding of the material” (p. 3). This highlights the need for educators to balance the use of AI tools with methods that ensure authentic student engagement and assessment.

While LLMs offer valuable opportunities for innovation in education, it is imperative that educators be aware of the ethical and practical considerations involved in prompt engineering. By integrating the above strategies, educators can promote equitable, accurate, and pedagogically sound applications of generative AI. This not only enriches the learning experience but also upholds the standards of equity and accuracy essential in education.

As Miao and Cukurova (2024) propose, colleges and universities should validate the trustworthiness of AI tools, establish regulatory safeguards for responsible use, and design contextualized performance-based assessments to evaluate teachers’ AI competencies. Additionally, teacher education programs should incorporate AI-focused professional development pathways that align with existing qualification standards and national priorities.

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ORCID

Ana-Paula Correia  <http://orcid.org/0000-0003-0806-7835>

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Additional resources for classrooms

1. Brand, S. (2023). Meet TRACI: User's guide to the TRACI prompt framework for ChatGPT, *StructuredPrompt.com*. <https://structuredprompt.com/wp-content/uploads/2023/08/TRACI-Users-Guide-v1.pdf>.

This document provides a detailed explanation of the TRACI prompt-development framework for ChatGPT and other LLM-based generative AI tools. An acronym for Task, Role, Audience, Create, and Intent, the TRACI framework helps users consider the most important parameters to include in an AI prompt, from the basic activity being requested to the overall purpose of the output, constraints or expectations related to the output, or the target audience for the content being generated

2. Mollick, L. & Mollick, E. (2024). Student Exercises. *More Useful Things: AI Resources*. <https://www.moreusefulthings.com/student-exercises>.

This resource provides an extensive collection of prompts designed specifically for AI tutoring applications, offering insights into crafting questions and exercises that maximize the educational potential of generative AI tools. This guide is structured around practical use cases, such as student engagement, personalized learning, and formative assessment, making it highly relevant for educators integrating AI into their teaching strategies. With examples and templates for various subject areas, the site emphasizes flexibility and creativity in adapting prompts to different learning objectives. The website is a valuable tool for educators at all levels, especially those exploring AI's role in fostering interactive and adaptive learning environments

3. Walter, Y. (2024). Embracing the future of Artificial Intelligence in the classroom: the relevance of AI literacy, prompt engineering, and critical thinking in modern education. *International Journal of Educational Technology in Higher Education*, 21, 15. <https://doi.org/10.1186/s41239-024-00448-3>.

This research article uses a case-study approach, paired with a narrative literature review, to examine the capabilities and limitations of artificial intelligence in the classroom. The researcher found that the careful use of genAI tools could allow for personalized learning, support the instructor in differentiating instruction, streamline grading and other administrative functions, and facilitate collaborative learning among students.